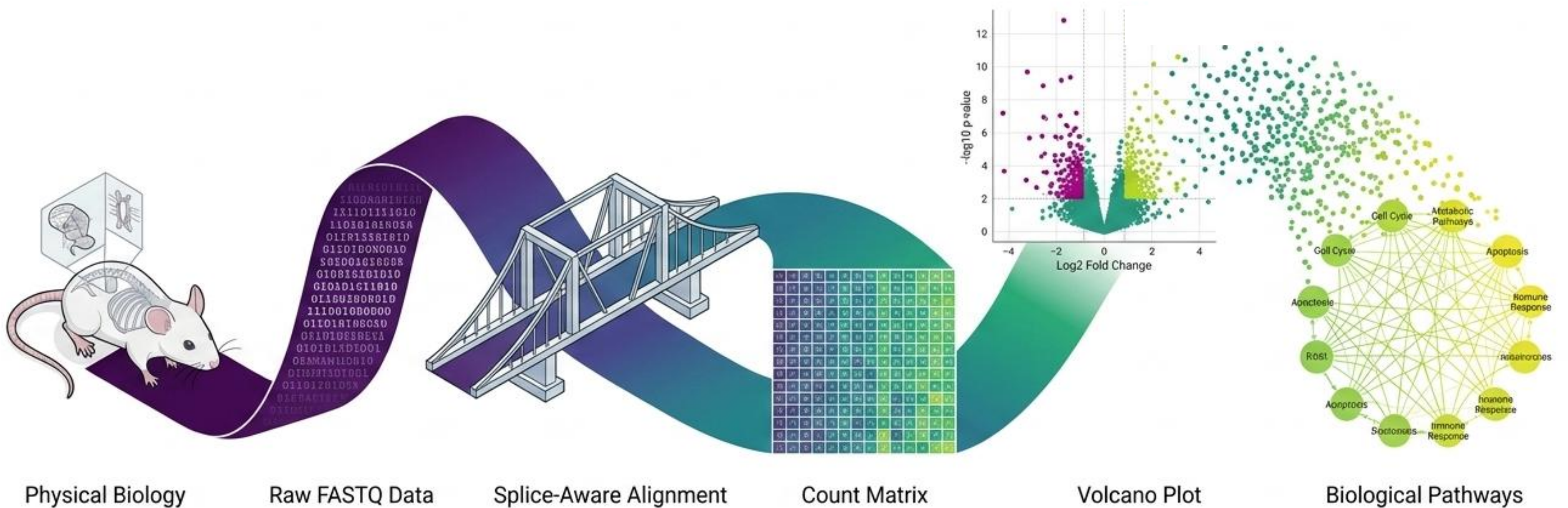
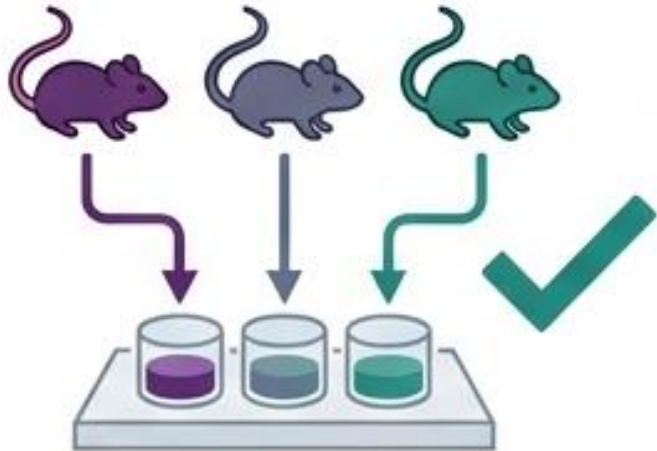
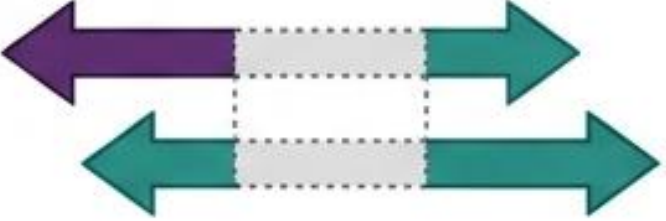
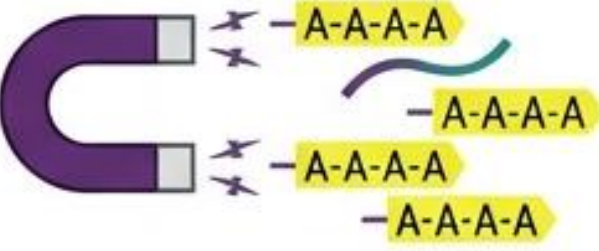


# RNA Sequencing and Analysis



|                                     | Technical Replicate                                                                                | Biological Replicate                                                                                |
|-------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Technical vs. Biological Replicates |                  |                  |
| Single-end vs. Paired-end           | Single-end<br> | Paired-end<br> |
| Poly-A vs. Ribo-Zero                | Poly-A<br>     | Ribo-Zero<br>  |

Garbage in, garbage out. No algorithm can rescue fully confounded batch effects.

### Statistical Power

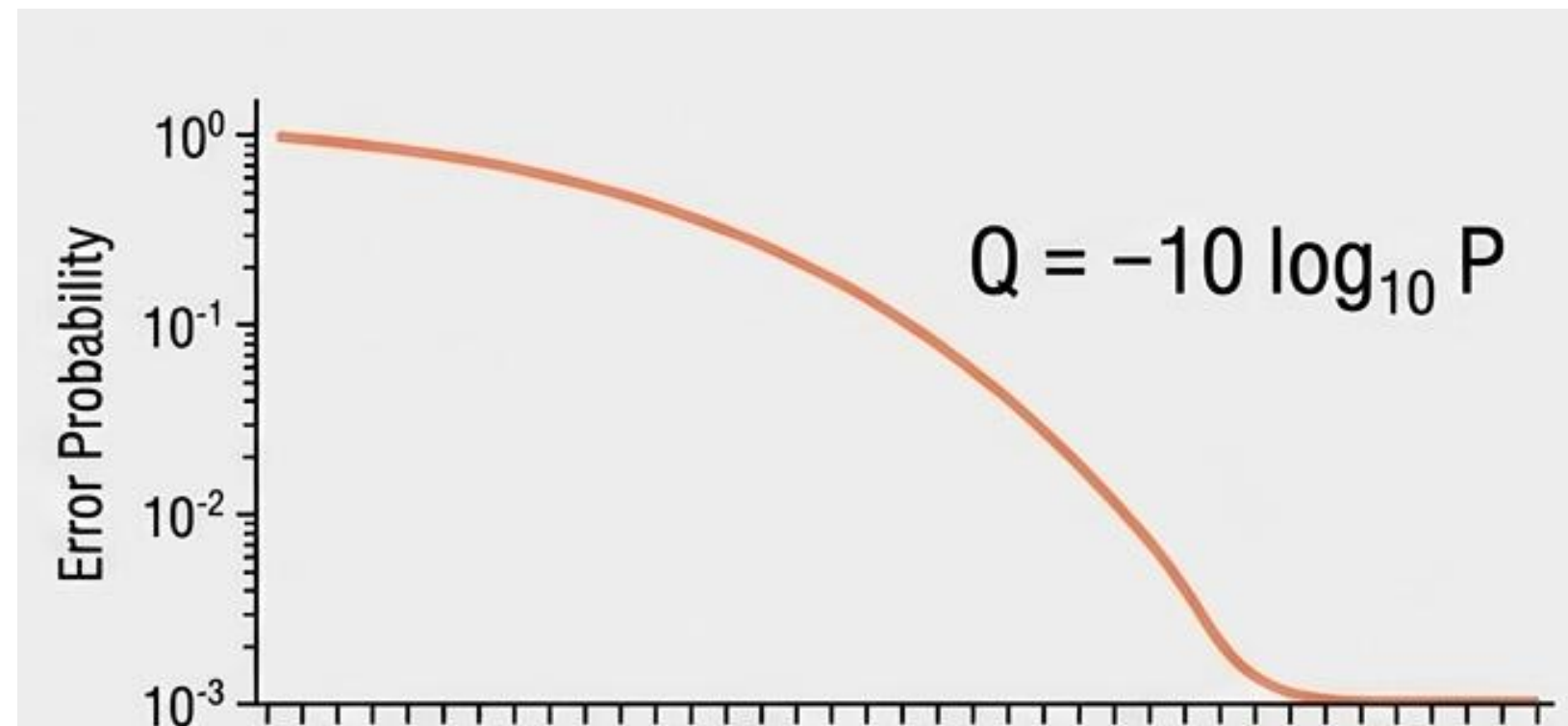
A minimum of 3 biological replicates is mandatory; highly heterogeneous tissues demand  $n \geq 5$ .

Confounding sequencing lanes with biological treatment groups. If all Control samples run on Lane 1 and all Treated on Lane 2, biological variance becomes mathematically indistinguishable from optical machine variance.



# FASTQ file structure

```
@SEQ_ID:1:1101:10000#0/1  ← Header & Read 1 ID
GATTGGGGTTCAAAGCAGTATCGATCAAATAGTAAATCCATTGTTC AAC  ← Base Sequence
+  ← Separator
!''*(((((***+))%%%++)(%%%%).1***-+*'))**55CCF>>>>>  ← Phred Quality Scores (ASCII)
```



## The Phred Scale

Quality scores are logarithmic. A score of Q20 implies a 1 in 100 probability of an incorrect base call (99% accuracy). Q30 is 1 in 1,000 (99.9% accuracy).

## Optical Artifacts

The sequencer relies on fluorescent flashes. As clusters degrade over subsequent chemical cycles, the signal-to-noise ratio drops, intrinsically causing quality to plummet at the 3' end of the read.

# Adapter trimming and QC

```
trim_galore \  
  --paired \  
  --quality 20 \  
  --length 36 \  
  --cores 16 \  
  --fastqc \  
  --output_dir $TRIM \  
  $R1 $R2
```

## Purpose

Remove artificial adapter read-through and low-quality 3' ends that prevent reads from mapping accurately to the reference genome.

## Parameter Tuning

--quality 20: Enforces a 99% accuracy threshold for base retention.

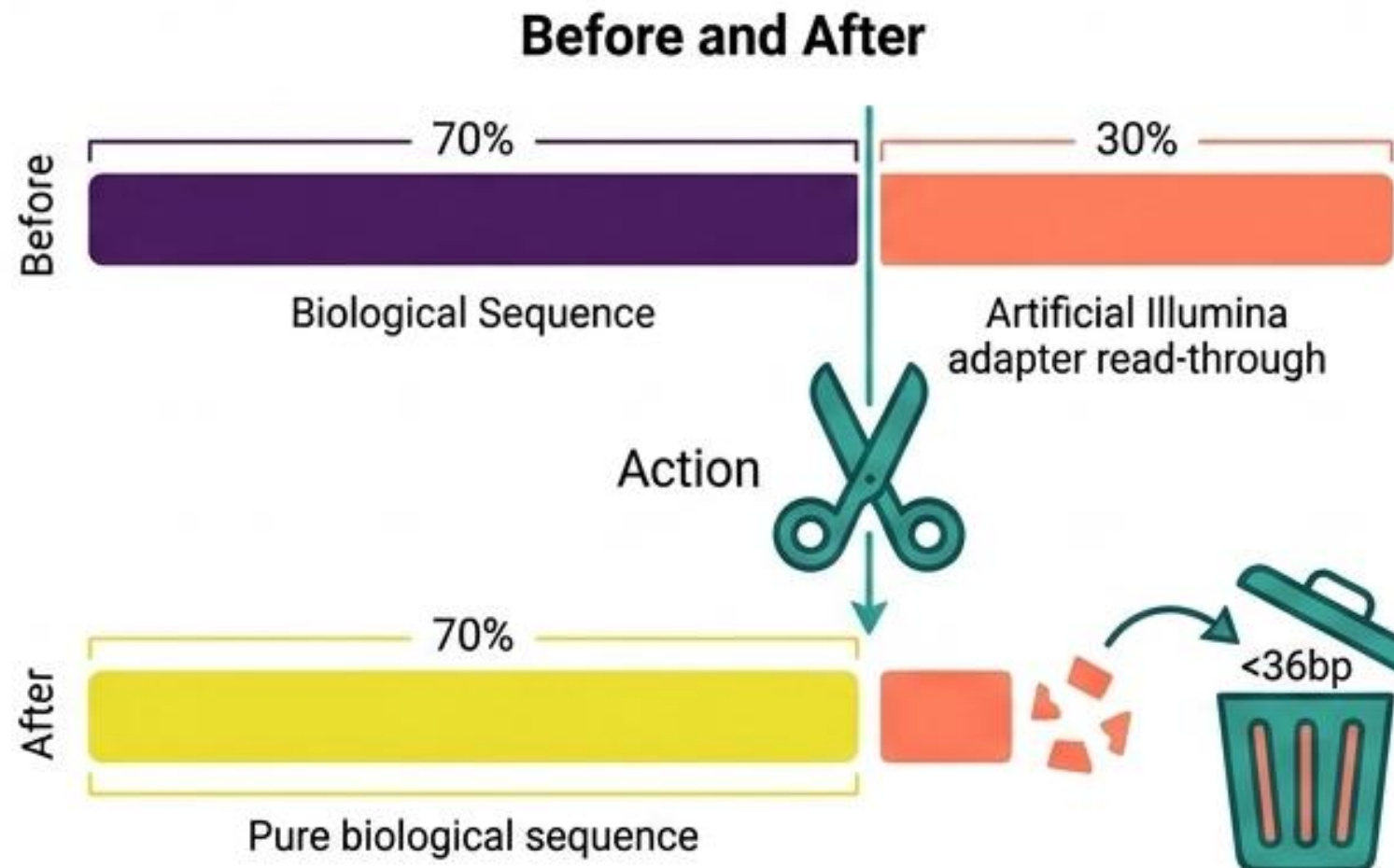
--length 36: Discards any read that becomes shorter than 36 bases after trimming. Reads shorter than this lose statistical uniqueness and map arbitrarily across the genome.

## Common Mistakes

Over-trimming. Setting quality thresholds too high (e.g., Q30) unnecessarily discards valid biological data, causing unique mapping rates to plummet.



## Cleavage Diagram



# Adapter trimming and QC

Per base sequence quality

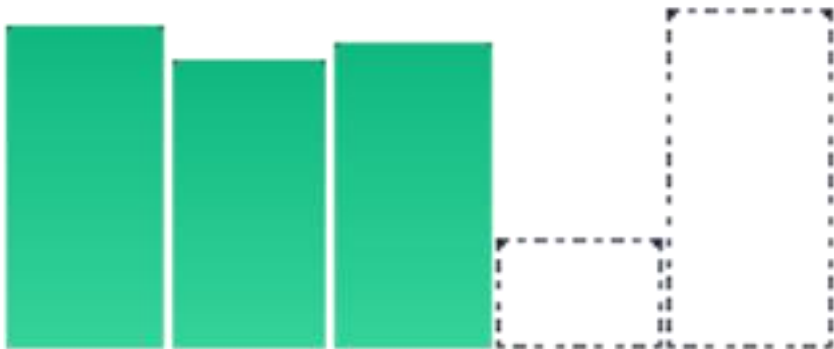


Position in read(bp) -> quality degrades at 3' end

Before trimming



After trimming

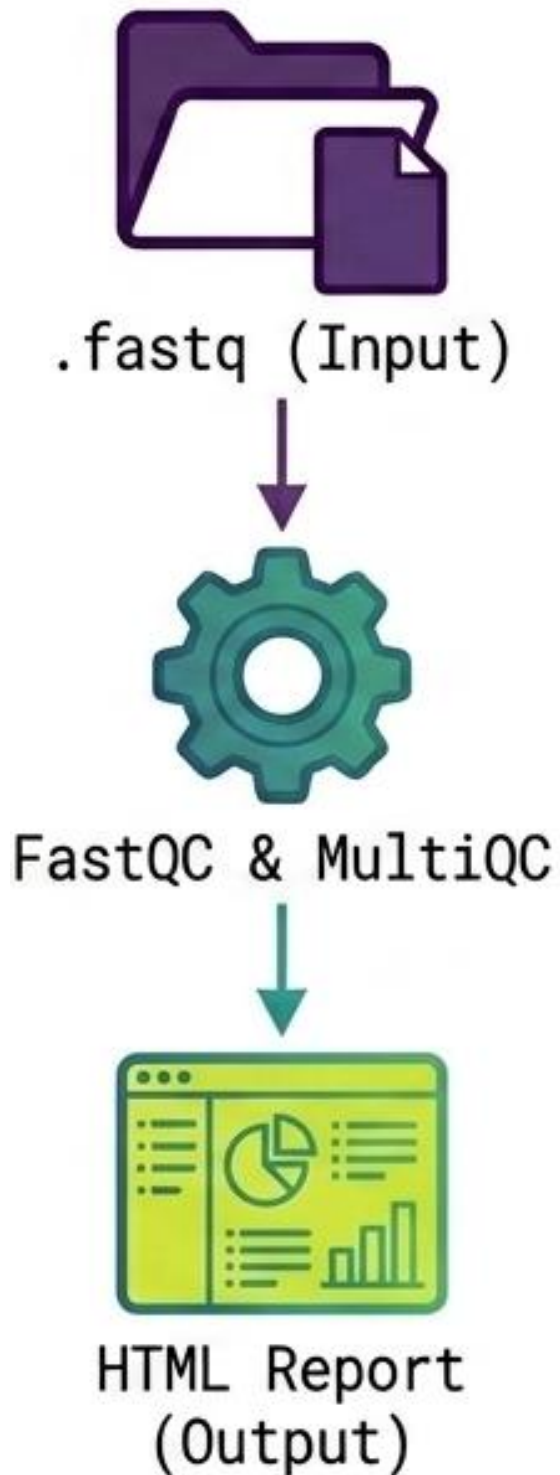


Diagnostic Comparison Matrix

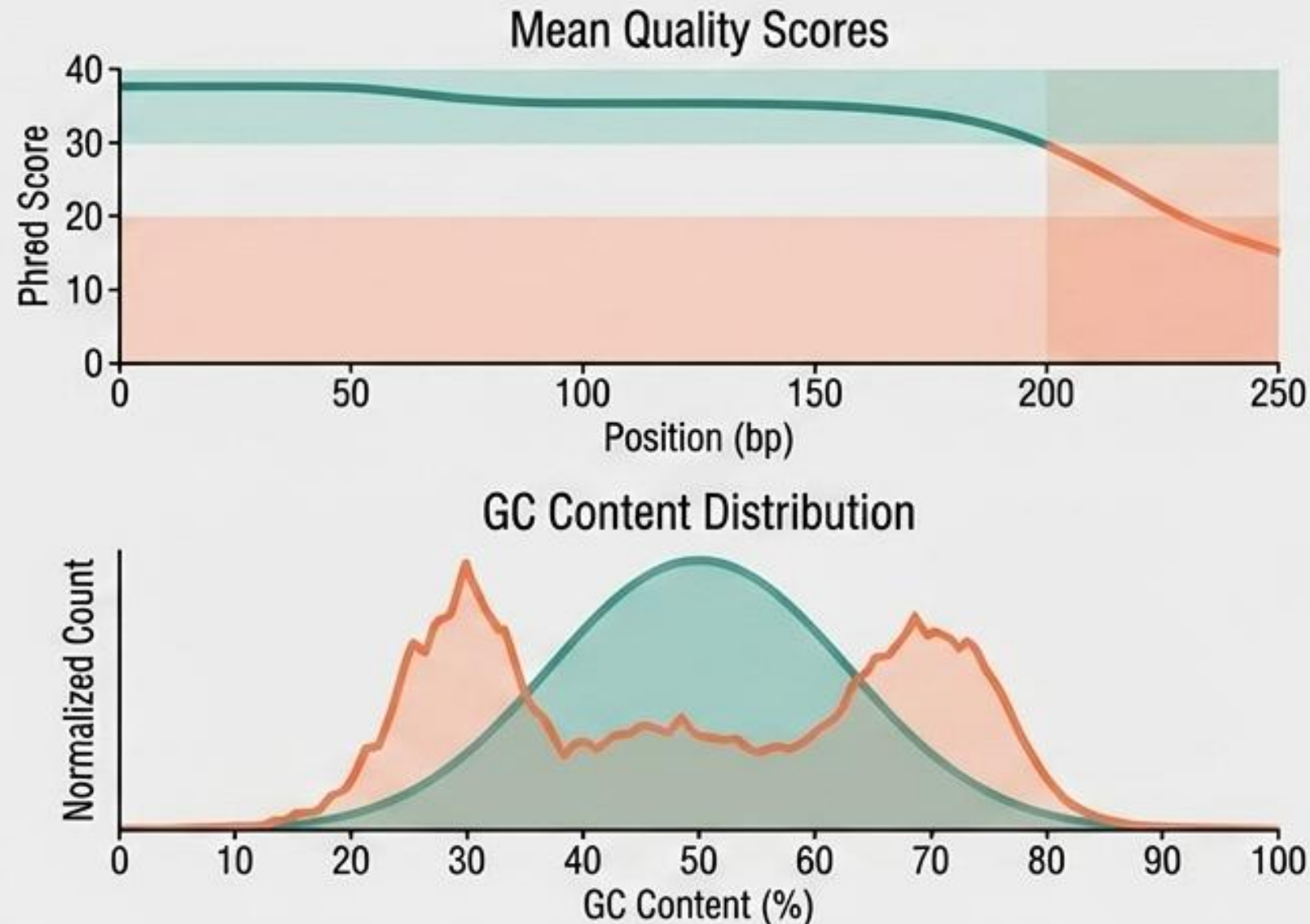
| Feature        | Trim Galore    | fastp              |
|----------------|----------------|--------------------|
| Engine         | Python wrapper | C++ binary         |
| Speed          | Slower         | Ultra-fast         |
| Multithreading | Limited        | Native support     |
| Integrated QC  | No             | Yes (HTML reports) |



# MultiQC



```
multiqc $PRE_QC $TRIM -o $QC/multiqc_pretrim \
--filename multiqc_pretrim.html -f
```



## Purpose

Aggregate individual FastQC reports into a single systemic overview to spot lane failures or global issues before investing computational resources.

## Data Flow

Parses individual .zip files  
-> **Aggregated HTML report.**

## Interpretation

A drop in 3' quality is a standard Illumina artifact, easily fixed by trimming. However, a **bi-modal GC content curve** is an **immediate red flag** suggesting library contamination (e.g., mycoplasma or adapter dimers).

# Reference Indexing



```
# Generate STAR genome index
STAR \
  --runMode genomeGenerate \
  --genomeDir $STAR_INDEX \
  --genomeFastaFiles $GENOME \
  --sjdbGTFfile $GTF \
  --sjdbOverhang 149 \
  --runThreadN $THREADS
```

- Purpose
- Aligners can not search the raw 3GB genome efficiently. so we have to create a optimized search index
- Key parameters
- sjdbOverhang should be read length -1

| gencode_v19_annotation_gtf[gencode_v19_annotation_gtf['gene_name']=='TERT'] |         |        |            |         |         |       |        |       |                    |                    |                |             |
|-----------------------------------------------------------------------------|---------|--------|------------|---------|---------|-------|--------|-------|--------------------|--------------------|----------------|-------------|
|                                                                             | seqname | source | feature    | start   | end     | score | strand | frame | gene_id            | transcript_id      | gene_type      | gene_status |
| 696990                                                                      | chr5    | HAVANA | gene       | 1253262 | 1295184 | NaN   | -      | 0     | ENSG00000164362.14 | ENSG00000164362.14 | protein_coding | KNOWN       |
| 696991                                                                      | chr5    | HAVANA | transcript | 1253262 | 1283125 | NaN   | -      | 0     | ENSG00000164362.14 | ENST00000484238.2  | protein_coding | KNOWN       |
| 696992                                                                      | chr5    | HAVANA | exon       | 1282544 | 1283125 | NaN   | -      | 0     | ENSG00000164362.14 | ENST00000484238.2  | protein_coding | KNOWN       |
| 696993                                                                      | chr5    | HAVANA | exon       | 1280273 | 1280453 | NaN   | -      | 0     | ENSG00000164362.14 | ENST00000484238.2  | protein_coding | KNOWN       |
| 696994                                                                      | chr5    | HAVANA | exon       | 1279406 | 1279585 | NaN   | -      | 0     | ENSG00000164362.14 | ENST00000484238.2  | protein_coding | KNOWN       |
| 696995                                                                      | chr5    | HAVANA | exon       | 1278756 | 1278911 | NaN   | -      | 0     | ENSG00000164362.14 | ENST00000484238.2  | protein_coding | KNOWN       |
| 696996                                                                      | chr5    | HAVANA | exon       | 1268635 | 1268748 | NaN   | -      | 0     | ENSG00000164362.14 | ENST00000484238.2  | protein_coding | KNOWN       |
| 696997                                                                      | chr5    | HAVANA | exon       | 1266579 | 1266650 | NaN   | -      | 0     | ENSG00000164362.14 | ENST00000484238.2  | protein_coding | KNOWN       |
| 696998                                                                      | chr5    | HAVANA | exon       | 1260589 | 1260715 | NaN   | -      | 0     | ENSG00000164362.14 | ENST00000484238.2  | protein_coding | KNOWN       |
| 696999                                                                      | chr5    | HAVANA | exon       | 1258713 | 1258774 | NaN   | -      | 0     | ENSG00000164362.14 | ENST00000484238.2  | protein_coding | KNOWN       |
| 697000                                                                      | chr5    | HAVANA | exon       | 1255402 | 1255526 | NaN   | -      | 0     | ENSG00000164362.14 | ENST00000484238.2  | protein_coding | KNOWN       |

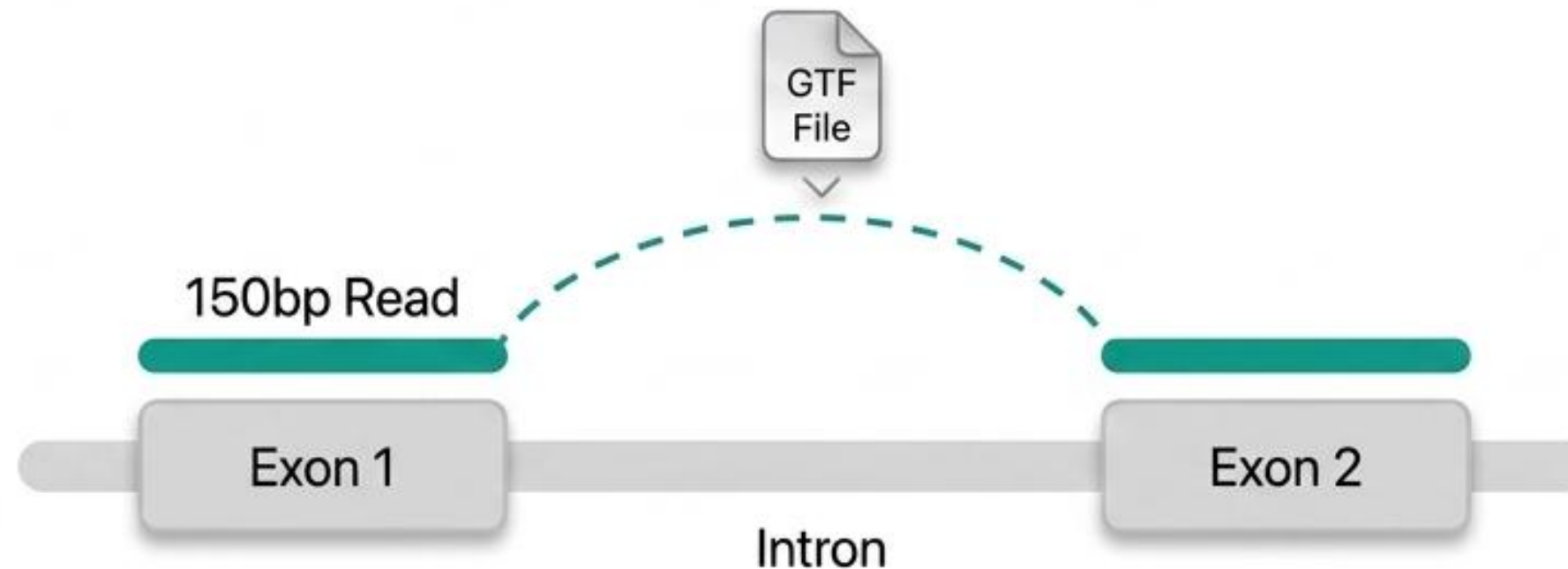


# Quantification



```
STAR --runThreadN $THREADS \
--genomeDir $STAR_INDEX \
--readFilesIn $R1 $R2 \
--outSAMtype BAM SortedByCoordinate \
--sjdbGTFfile $GTF_FILE \
--quantMode GeneCounts \
--alignIntronMax 1000000
```

## Splice-Aware Visualization



### Purpose

- Map biological reads back to their precise genomic coordinates. STAR natively models RNA splicing mechanisms.

### Key Parameters

- **--sjdbGTFfile:** Informs the aligner of biologically known splice junctions, anchoring the search across massive intronic gaps.
- **--quantMode GeneCounts:** Generates ReadsPerGene.out.tab simultaneously with alignment, saving an entire computational step later.

### Outputs

- A massive, coordinate-sorted BAM file containing exact positional data (Viridis Yellow).





| Quantification Paradigms                                                                                                                                                                                               |                                                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| featureCounts<br>Alignment-Based                                                                                                                                                                                       | Salmon<br>Alignment-Free                                                                                                                                                                                           |
| <ul style="list-style-type: none"><li>• Requires heavy BAM files.</li><li>• Counts reads physically overlapping genomic features.</li><li>• Slower, strict parameters.</li><li>• Outputs raw integer counts.</li></ul> | <ul style="list-style-type: none"><li>• Bypasses BAM files entirely.</li><li>• Uses k-mers to probabilistically estimate abundance.</li><li>• Blazing fast.</li><li>• Outputs TPM and effective lengths.</li></ul> |

# Quantification QC



.sam/.bam



Samtools / Qualimap

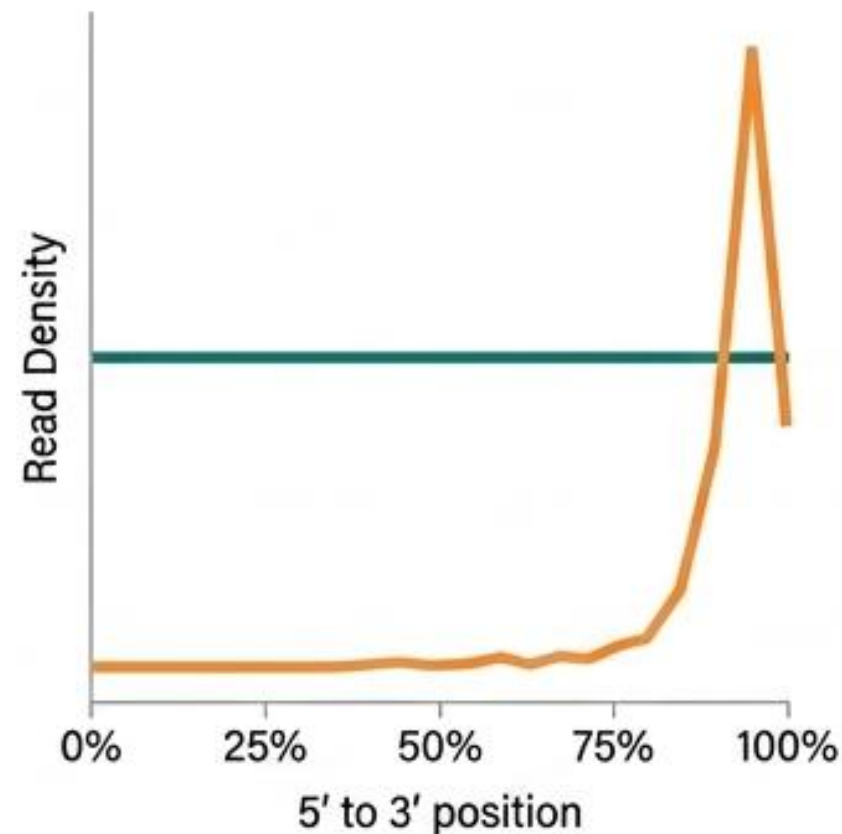


Sorted .bam /  
QC Metrics

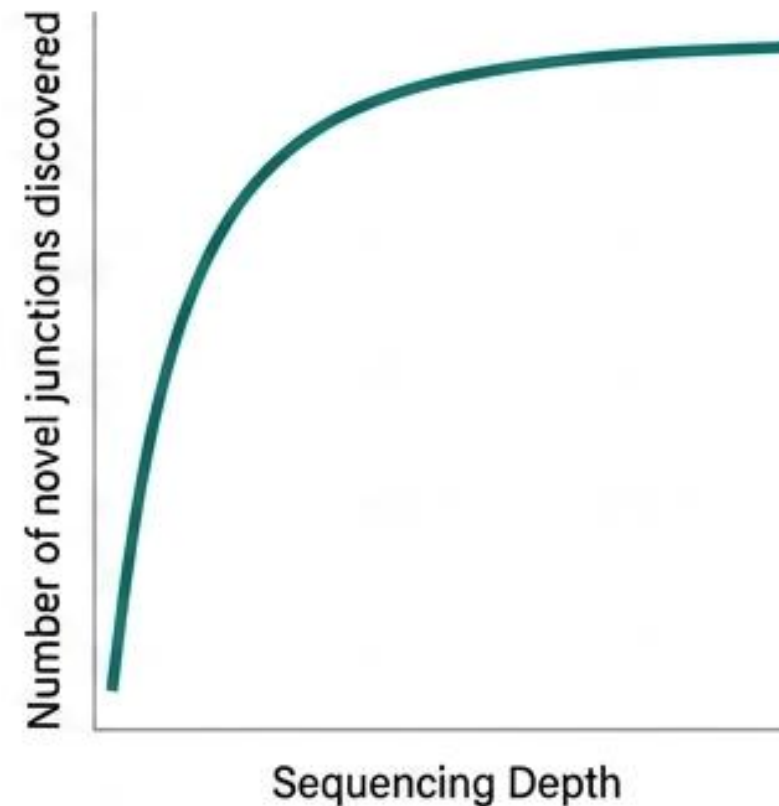
```
geneBody_coverage.py -r $BED12 -i $BAM -o $OUT  
samtools flagstat -@ $THREADS $BAM > flagstat.txt
```

## Diagnostic Dashboard

Gene Body Coverage



Junction Saturation



## Purpose

Interrogate the BAM file to assess library integrity and alignment efficiency before investing in quantification.

## Flagstat Interpretation

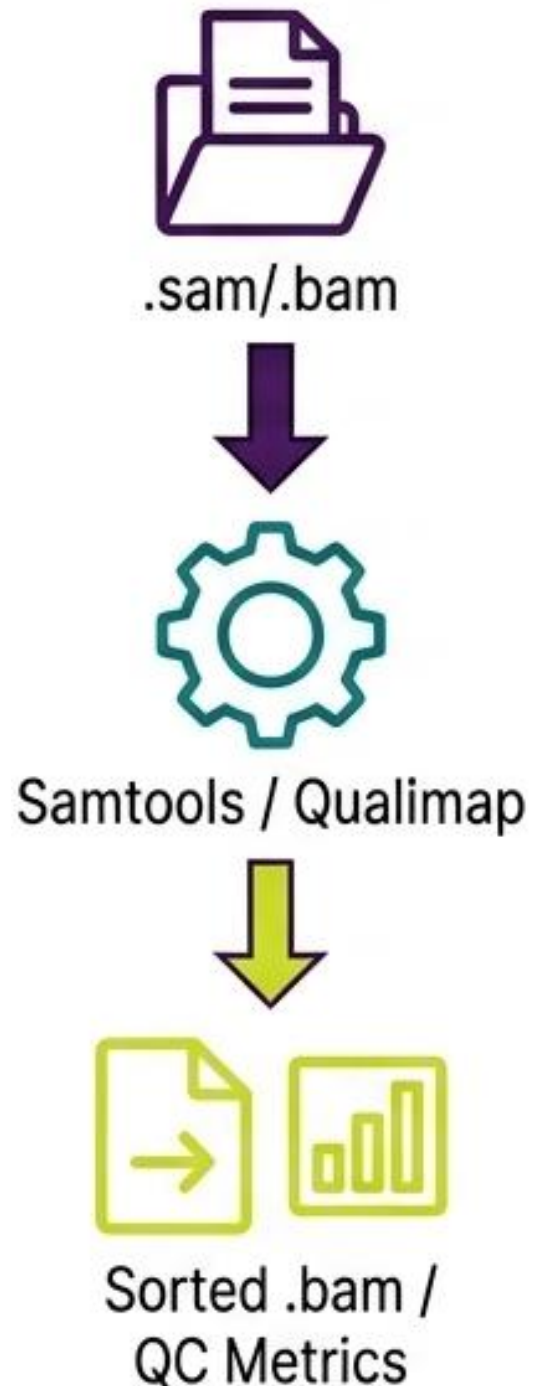
`samtools flagstat` provides the global mapping rate. >80% uniquely mapped reads is the benchmark for high-quality RNA-seq.

## Degradation Check

`geneBody_coverage.py` is the ultimate arbiter of RNA quality. Because mRNA degrades from the 5' end, heavy 3' coverage bias indicates the physical RNA was fragmented prior to sequencing, heavily skewing abundance estimates.



# Quantification QC



```
samtools index sorted.bam
```

```
qualimap rnaseq -bam sorted.bam \  
            -gtf annotation.gtf \  
            -outdir qualimap_out
```

```
infer_experiment.py -r annotation.bed \  
                    -i sorted.bam
```

## Purpose

### ⚠ Strandedness (RSeQC)

Did the library preparation preserve directionality? Ensures correct assignment of overlapping genes.

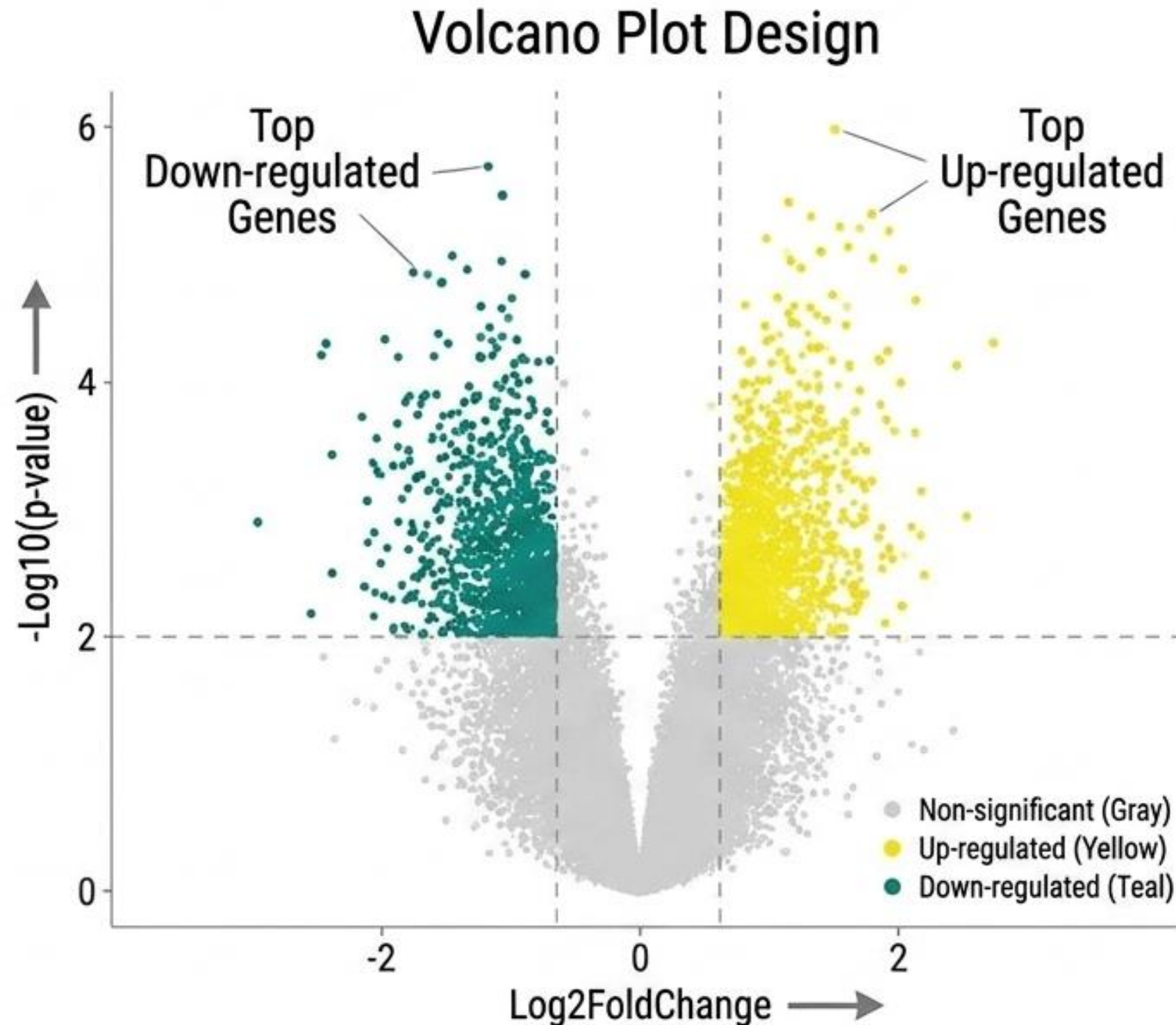
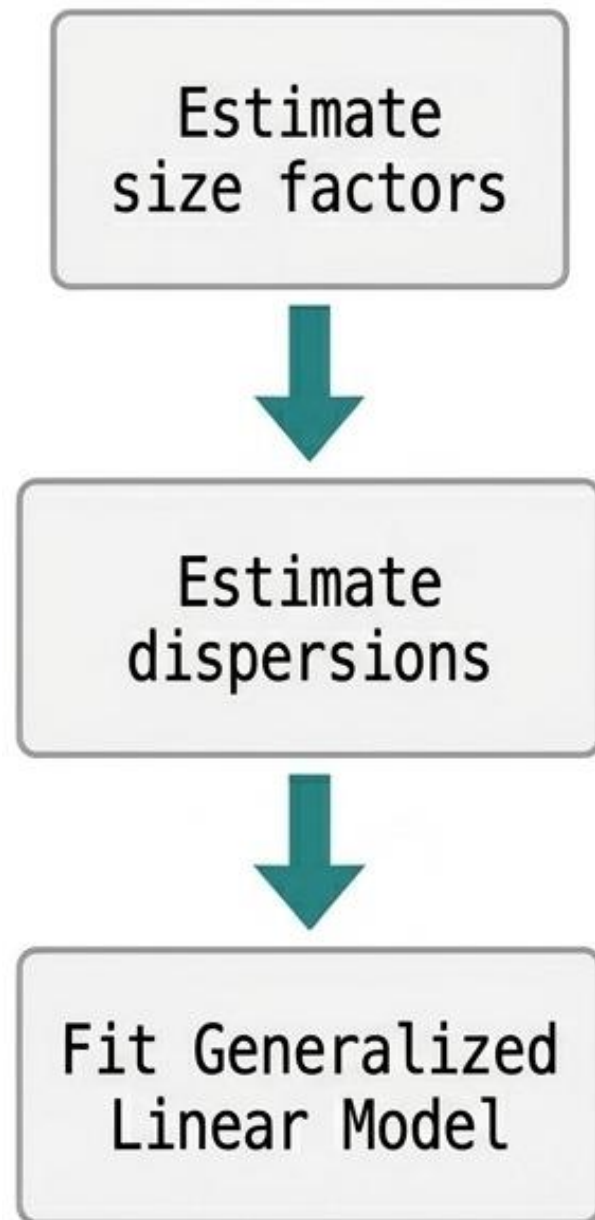
### ⚠ Uniform Coverage (Qualimap)

Are reads evenly distributed? Checks for severe 3' or 5' transcript degradation biases.

### ⚠ Genomic Distribution

Are reads properly mapping to exons, or falling into unannotated intronic/intergenic space?

# Differential Expression Analysis (DESeq2)



## Purpose

- Deploy statistical testing (e.g., DESeq2, edgeR) to distinguish true biological changes from technical noise.

## Statistical Nuance

- Differential expression tools demand raw, un-normalized counts (not TPM or FPKM). Their underlying mathematical models (Negative Binomial distribution) require raw variance estimates to calculate accurate dispersion metrics.

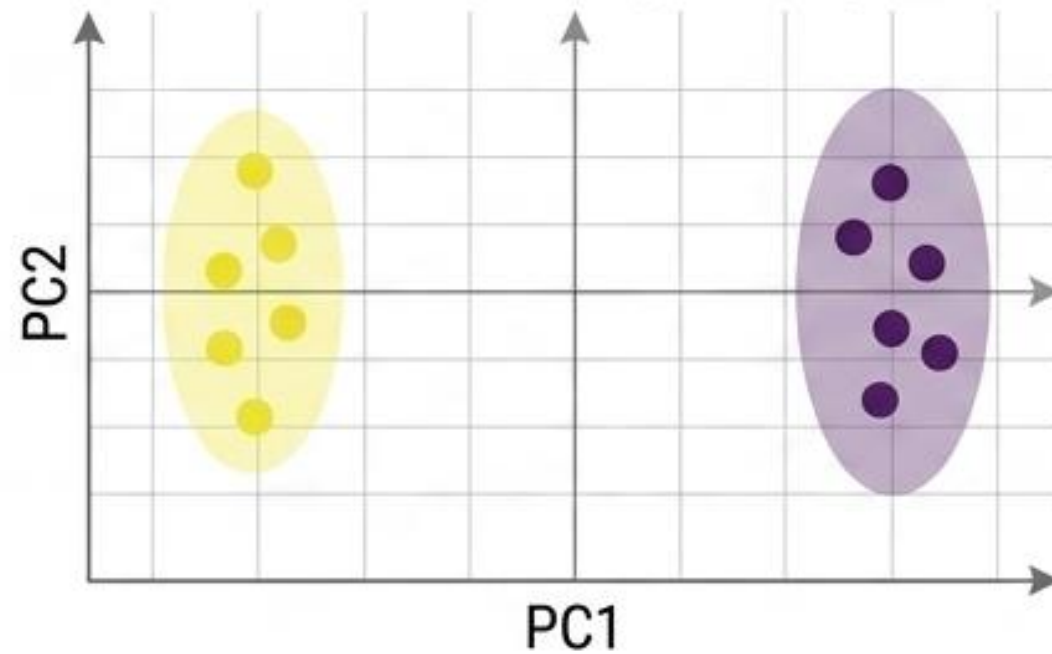
## Analytical Mistakes

- Relying exclusively on arbitrary p-value cutoffs without evaluating the magnitude of the change (Fold Change) or the baseline expression level (Base Mean) of the gene.

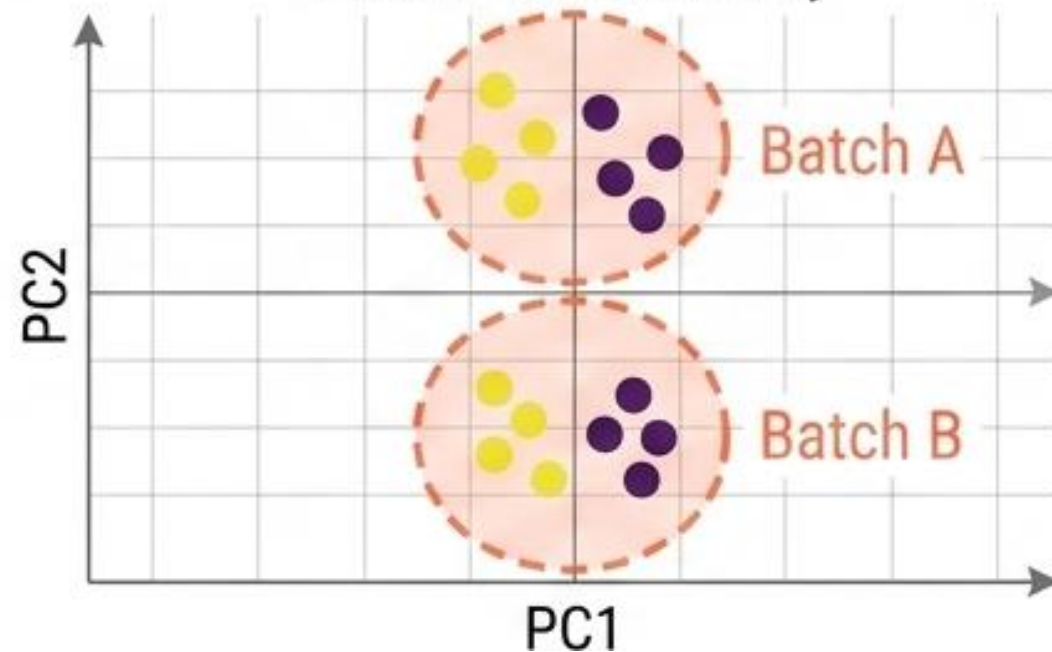


# PCA Plots

Ideal Scenario: Biological Separation



Batch Effect Reality



## Purpose

Perform an unbiased dimensionality reduction (**Principal Component Analysis**) to view the **global topography** of the entire dataset.

## Interpretation

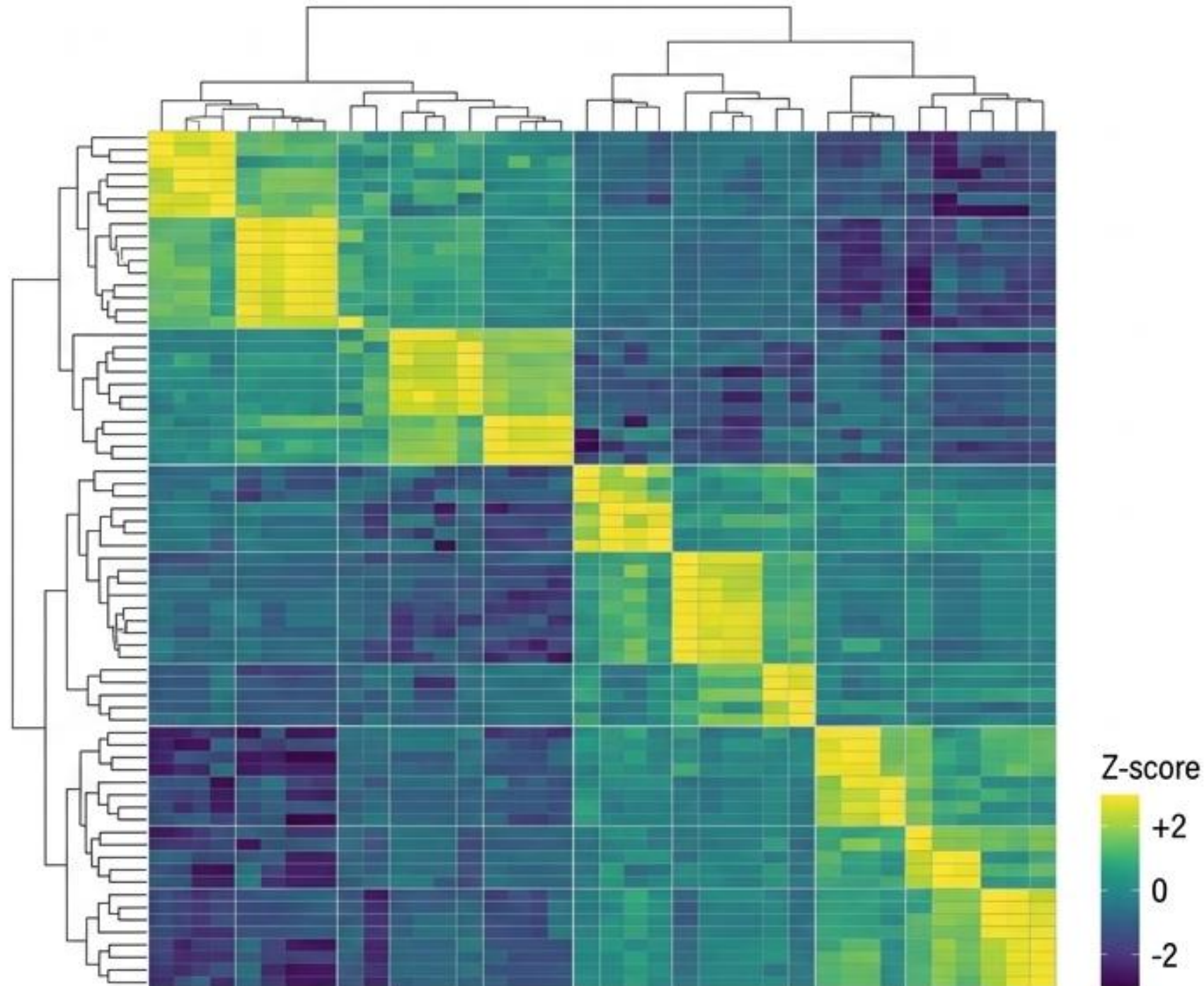
PC1 captures the maximum axis of variance. In a successful experiment, PC1 should map cleanly to the primary biological variable.

## Covariate Correction

If PCA reveals that samples cluster primarily by **sequencing batch**, **extraction date**, or **sex** (when not the variable of interest), these **hidden variables** must be **mathematically modeled and subtracted** (e.g., using **ComBat-Seq**) before trusting the results.

# Expression Pattern Visualization

## Clustered Heatmap Design



### Purpose

- Visualize complex groups of genes behaving synchronously across different experimental conditions.

### Methodological Necessity

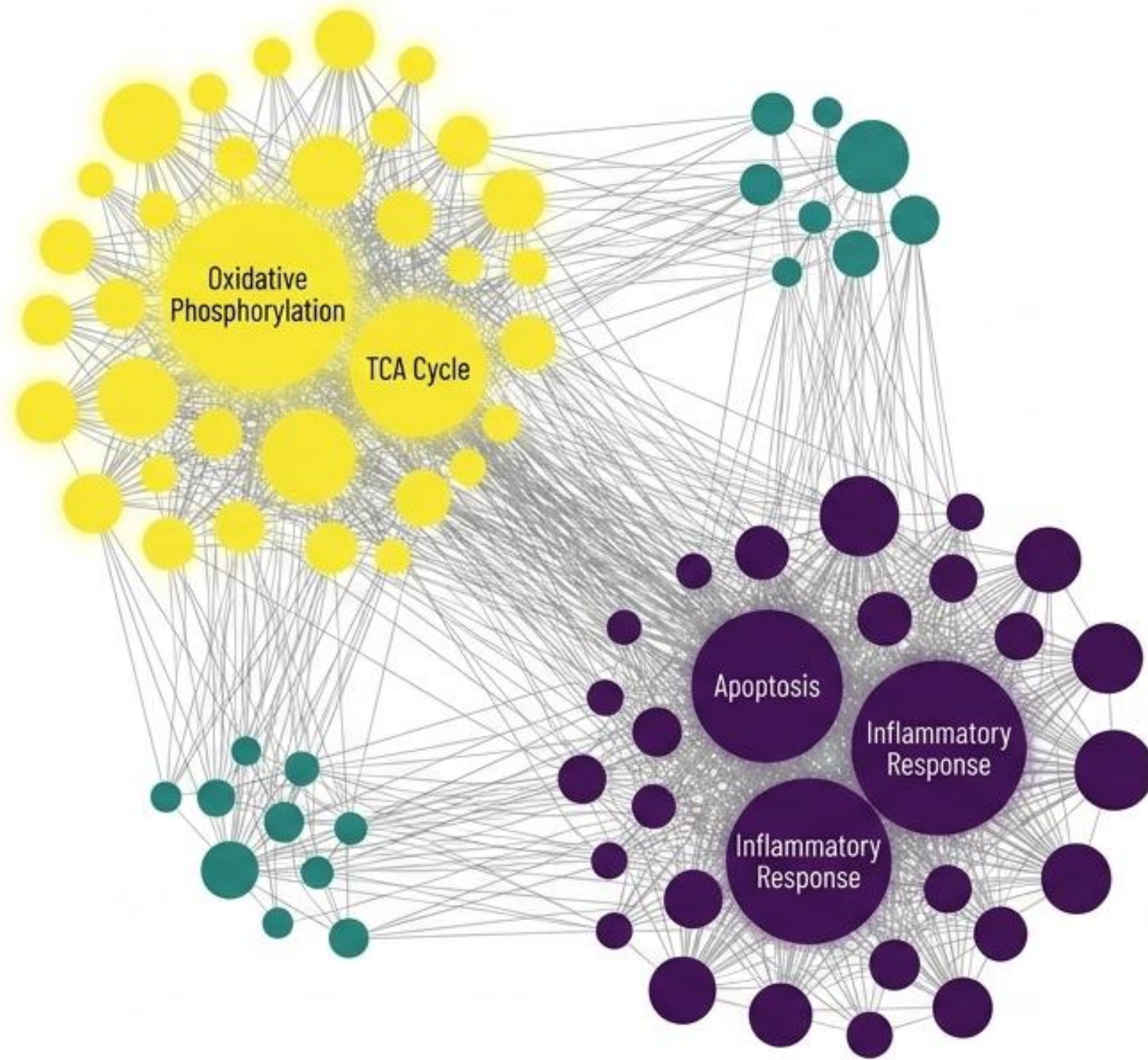
- Expression counts span orders of magnitude. A **Variance Stabilizing Transformation (VST)** is applied, followed by **row-scaling (Z-scores)**. This ensures highly expressed structural genes do not visually overwhelm subtle, yet critical, transcription factor changes.

### Biological Insight

- **Guilt by association.** Genes that cluster tightly together on the dendrogram often share regulatory promoter elements or operate within the same physical protein complex.



# Functional & Pathway Enrichment



## Purpose

Translate long lists of arbitrary gene symbols into recognizable biological mechanisms and phenotypes.

## Methodology

Gene Set Enrichment Analysis (GSEA).

## GSEA vs. ORA

Over-representation Analysis (ORA) relies on a rigid, arbitrary p-value cutoff, ignoring genes that fall just below the threshold.

GSEA evaluates the ENTIRE ranked list of genes without cutoffs, proving vastly more sensitive to small but coordinated expression shifts across an entire molecular pathway.